

ECE 607

Homework #1

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1)

$$f(x_1, x_2) = 3 + x_1^2 + x_2^2$$

$$g(x_1, x_2) = x_1 + x_2 + 1 = 0$$

$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda g(x_1, x_2)$$

$$\mathcal{L}(x_1, x_2, \lambda) = 3 + x_1^2 + x_2^2 + \lambda(x_1 + x_2 + 1)$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 + \lambda = 0$$

$$2x_1 = -\lambda$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 + \lambda = 0$$

$$2x_2 = -\lambda$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = x_1 + x_2 + 1 = 0$$

$$x_1 + x_2 = -1$$

$$x_1 = x_2$$

$$x_1 + x_1 = -1$$

$$2x_1 = -1$$

$$x_1 = -1/2$$

$$x_2 = -1/2$$

$$2x_1 = -\lambda$$

$$2(-1/2) = -\lambda$$

$$-2/2 = -\lambda$$

$$\lambda = 1$$

$$f(\overset{-1/2}{\cancel{x_1}}, \overset{-1/2}{\cancel{x_2}}) = 3 + (-1/2)^2 + (-1/2)^2$$

$$= 3 + 1/4 + 1/4$$

$$= 3 + 1/2$$

$$\frac{6}{2} + \frac{1}{2} = \frac{7}{2}$$

$$\boxed{7/2}$$

$$\boxed{(-1/2, -1/2, 7/2)}$$

2)

$$U(x|a,b) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$\int_{-\infty}^{\infty} U(x|a,b) dx = \int_a^b \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} [x]_a^b = \frac{b-a}{b-a} = \boxed{1}$$

$$E(x) = \int_{-\infty}^{\infty} x \cdot U(x|a,b) dx = \int_a^b x \cdot \frac{1}{b-a} dx$$

$$= \cancel{\frac{1}{b-a}} = \frac{1}{b-a} \int_a^b x = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b$$

$$= \frac{b^2 - a^2}{2} \cdot \frac{1}{b-a} = \frac{(b-a)(a+b)}{2(b-a)}$$

$$= \boxed{\frac{a+b}{2}}$$

$$\text{VAR}(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b$$

$$= \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3}$$

$$\frac{(b-a)(b^2 + ab + a^2)}{3(b-a)}$$

$$= \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2} \right)^2$$

$$= \frac{b^2 + ab + a^2}{3} - \frac{a^2 + 2ab + b^2}{4}$$

$$= \frac{4(b^2 + ab + a^2)}{12} - \frac{3(a^2 + 2ab + b^2)}{12}$$

$$= \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12}$$

$$= \frac{b^2 - 2ab + a^2}{12}$$

$$\text{VAR}(X) = \boxed{\frac{(b-a)^2}{12}}$$

3)

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\ln P(x) = \ln\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{(x-\mu)^2}{2\sigma^2}$$

$$\ln P(x) = -\ln(\sigma\sqrt{2\pi}) - \frac{1}{2\sigma^2}(x-\mu)^2$$

$$\frac{d}{dx} [\ln P(x)] = \frac{d}{dx} \left[-\ln(\sigma\sqrt{2\pi}) - \frac{1}{2\sigma^2}(x-\mu)^2 \right]$$

$$= 0 - \frac{1}{2\sigma^2} \cdot 2(x-\mu) = -\frac{x-\mu}{\sigma^2}$$

$$-\frac{x-\mu}{\sigma^2} = 0$$

$$-x + \mu = 0$$

$$-x = -\mu$$

$$\boxed{x = \mu}$$

4)

1) Yields the mean

2) Yields the median

3) Yields the mode